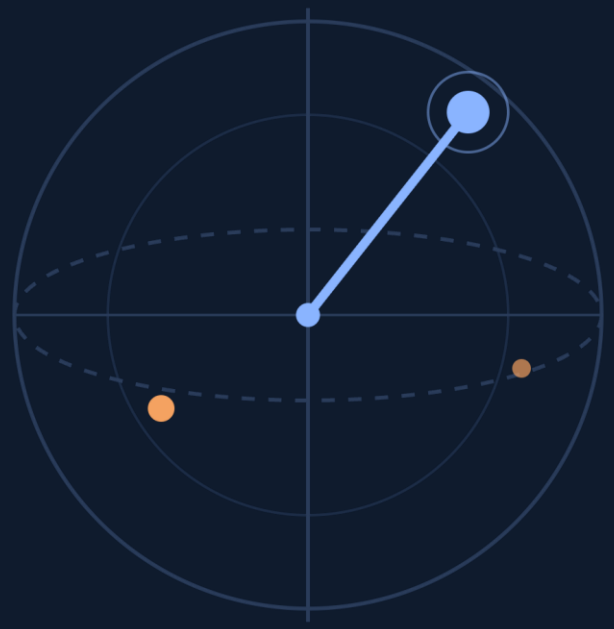
QISKIT FALL FEST 2026 · UOTTAWA · PROMPT 07

Learning molecular dipoles from fewer labels

Quantum kernel regression vs classical models on QM9



Team Honey Pack Research · Roman Daher · Douglass Jeffrey · Jadon Xia · October 2026

“How much reference data does a model need to predict molecular polarity?”

The dipole moment |μ| measures how far a molecule’s positive and negative charge are pulled apart; it drives solubility, intermolecular forces and spectroscopy. Every label in QM9 is a density-functional-theory calculation, so a model that needs fewer labels saves real compute. Quantum kernels are often proposed for exactly this small-data regime.

01

How fast does error fall as labels grow?

02

Can a quantum model match its classical twin?

03

Do models extrapolate to formulas never seen in training?

04

Does it survive shots, noise and real hardware?

Predict how polar a molecule is from its structure

INPUTS, FROM QM9

Atoms and their 3D positions

Z: the atomic number of every atom (H, C, N, O or F)

R: the x, y, z coordinates of every atom, in ångström

3 to 29 atoms per molecule, up to 9 of them heavier than hydrogen

Regression model

quantum or classical

OUTPUT

Dipole moment magnitude |μ|, in debye (D)

Charge separation: how much charge is pulled apart, times how far

Methane 0 D · water 1.85 D · QM9 average 2.7 D

Each QM9 label took a quantum-chemistry (DFT) calculation, so a model that needs fewer labels saves real compute. |μ| does not change when a molecule is rotated or moved, so neither may the model.

How do we evaluate the models?

TRACK A

Best Performing Classical Model?

How far can classical go from atoms and geometry?

- Up to all 99,198 training labels
- All 188 features
- Ridge, RBF kernel ridge, XGBoost, and a latent-charge network

TRACK B

Can we design a useful Quantum Model?

How does Quantum perform under the same inputs?

- At most 1,000 labels (N = 100, 300, 1,000)
- The same 10 inputs, one per qubit
- Same scaling, target, CV folds and 88-candidate grid

DATA

QM9: 133,885 molecules

Published QM9 (figshare collection 978904)	**133,885**
Uncharacterized: the optimized geometry no longer matches the intended molecule	**– 3,054**
Hard to converge, flagged in QM9's readme (11 listed, 3 already removed)	**– 8**
Duplicates: the same molecule listed twice; we keep one copy so no molecule sits in two sets	**– 140**
Final cleaned dataset	**130,683**

Ramakrishnan et al., Sci. Data 1, 140022 (2014) · all files verified against figshare's MD5 checksums · 0 parse failures

DATA

QM9, split by whole formulas

130,683

small organic molecules (up to 9 heavy atoms: C, N, O, F) with DFT geometries and dipoles

What's in it?

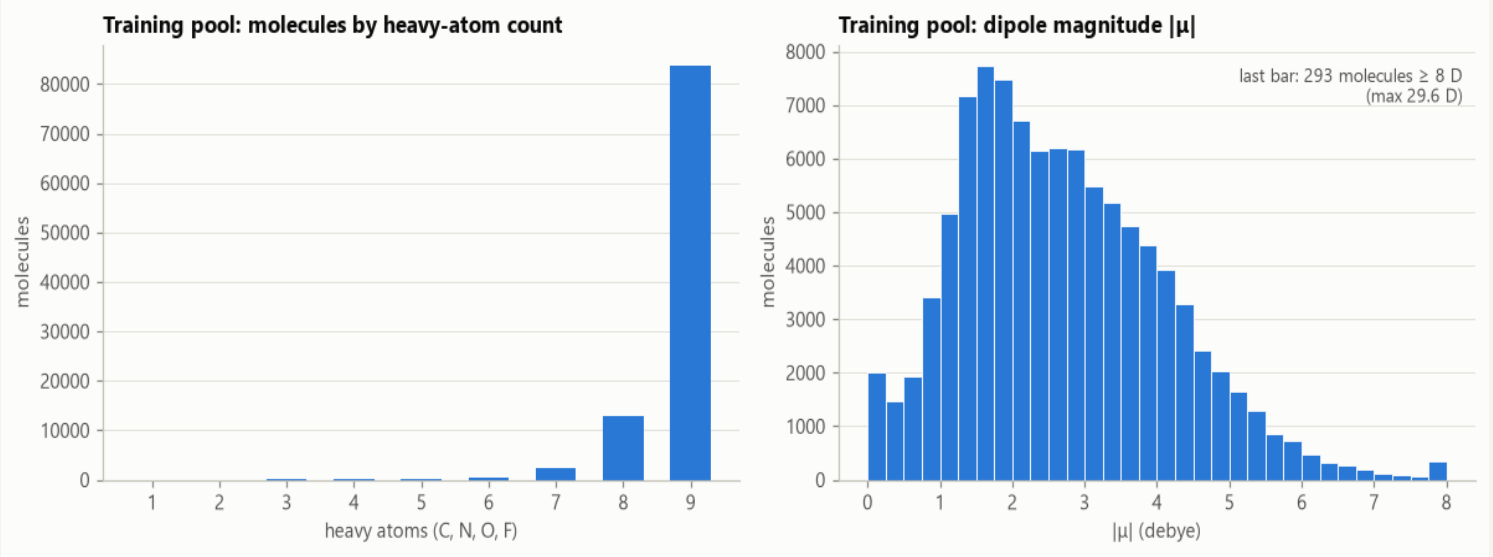
Small organic molecules from the GDB-17 enumeration: C, H, N, O and F only

Where labels come from?

DFT calculations (B3LYP/6-31G(2df,p)), not lab measurements

Any extreme outliers?

Rare zwitterions up to 29.6 D, still valid in our model



Set	Molecules	Formulas	Role
Training pool	99,198	497	General pool for training models
dev	5,000	257	Model development
dev_unseen	4,183	26	Development, held-out formulas
test_familiar	5,484	257	Test: formulas seen in training
test_unseen	16,818	93	Test: formulas never seen

CONSTRAINTS

Why we left QM9's DFT outputs out of our training features

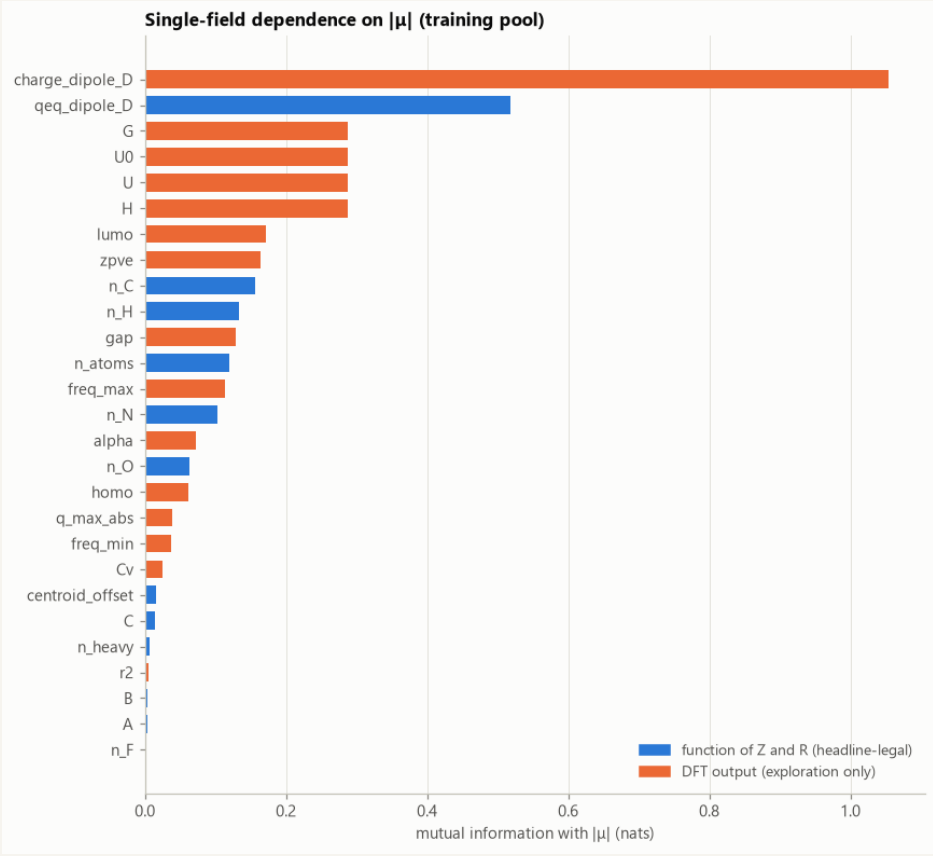
Circular at prediction time

To get them for a new molecule you would run DFT, and that run already gives |μ|. The model would predict nothing new.

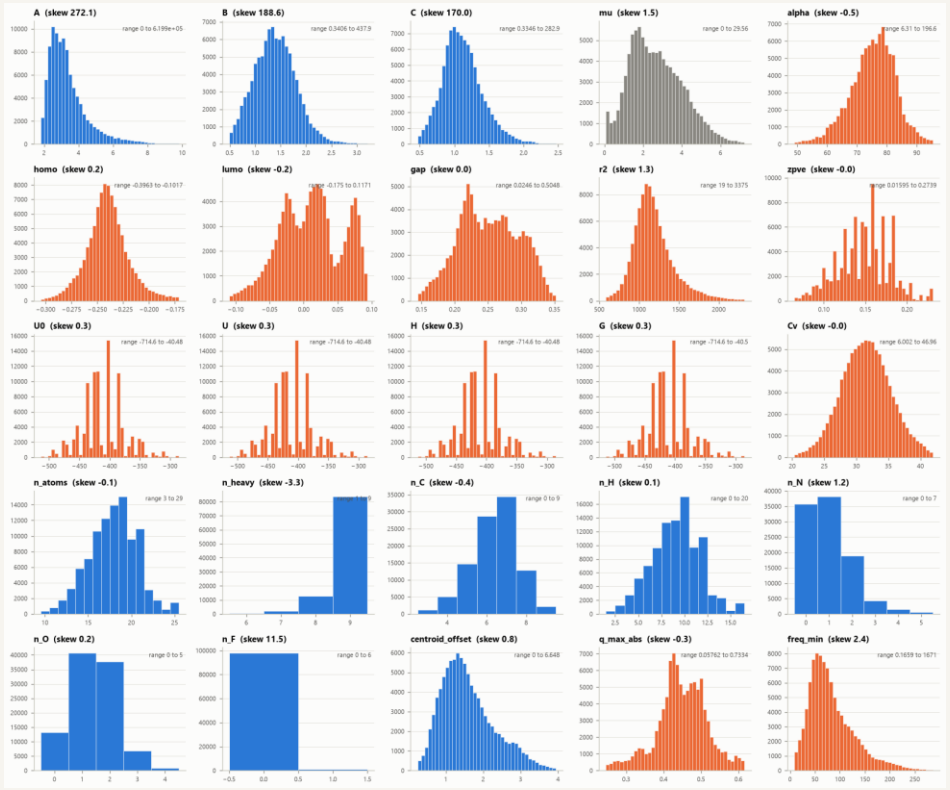
They leak the answer

The Mulliken point-charge dipole tracks |μ| at r = 0.94 (MI 1.05), twice as informative as our best legal feature.

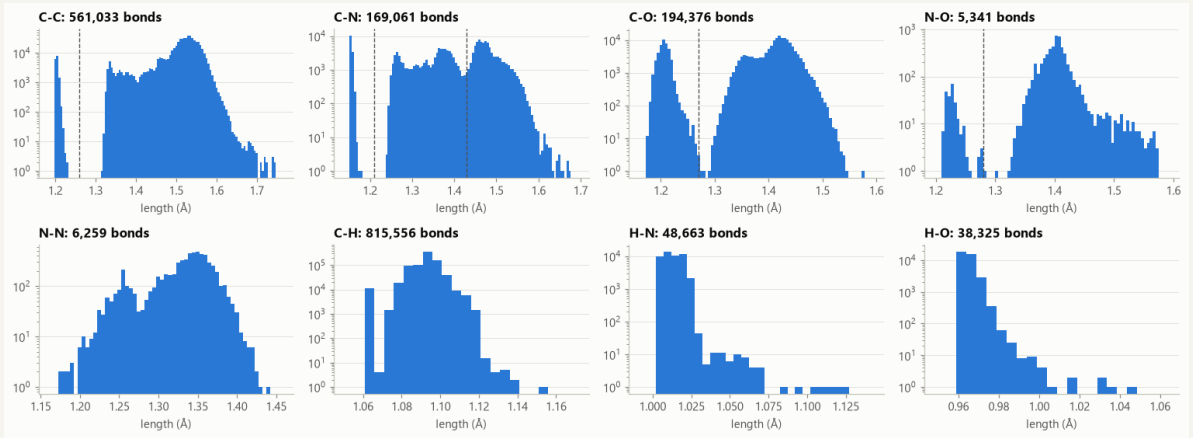
column	per	unit	meaning	role	dtype
id	molecule		QM9 index (GDB-17 numbering)	identifier / bookkeeping	int64
formula	molecule		element counts, e.g. C2H6O (from Z)	function of Z and R (headline-legal)	str
n_atoms	molecule		number of atoms (from Z)	function of Z and R (headline-legal)	int64
n_heavy	molecule		number of C, N, O, F atoms (from Z)	function of Z and R (headline-legal)	int64
Z	atom		atomic number	headline input	ndarray
R	atom	Å	Cartesian coordinates of the B3LYP-optimized geometry	headline input	ndarray
q	atom	e	Mulliken partial charge	DFT output (exploration only)	ndarray
A	molecule	GHz	rotational constant (largest)	function of Z and R (headline-legal)	float64
B	molecule	GHz	rotational constant	function of Z and R (headline-legal)	float64
C	molecule	GHz	rotational constant (smallest)	function of Z and R (headline-legal)	float64
mu	molecule	D	dipole moment magnitude |μ|	target	float64
alpha	molecule	$bohr^3$	isotropic polarizability	DFT output (exploration only)	float64
homo	molecule	Ha	highest occupied orbital energy	DFT output (exploration only)	float64
lumo	molecule	Ha	lowest unoccupied orbital energy	DFT output (exploration only)	float64
gap	molecule	Ha	lumo − homo	DFT output (exploration only)	float64
r2	molecule	$bohr^2$	electronic spatial extent (R^2)	DFT output (exploration only)	float64
zpve	molecule	Ha	zero-point vibrational energy	DFT output (exploration only)	float64
U0	molecule	Ha	internal energy at 0 K	DFT output (exploration only)	float64
U	molecule	Ha	internal energy at 298.15 K	DFT output (exploration only)	float64
H	molecule	Ha	enthalpy at 298.15 K	DFT output (exploration only)	float64
G	molecule	Ha	free energy at 298.15 K	DFT output (exploration only)	float64
Cv	molecule	cal/(mol·K)	heat capacity at 298.15 K	DFT output (exploration only)	float64
freqs	mode	cm^{-1}	harmonic vibrational frequencies (3n − 6 modes)	DFT output (exploration only)	ndarray
smiles	molecule		SMILES of the relaxed geometry	identifier / bookkeeping	str
smiles_gdb	molecule		SMILES of the GDB-17 input structure	identifier / bookkeeping	str
inchi	molecule		InChI of the relaxed geometry	identifier / bookkeeping	str
inchi_gdb	molecule		InChI of the GDB-17 input structure	identifier / bookkeeping	str

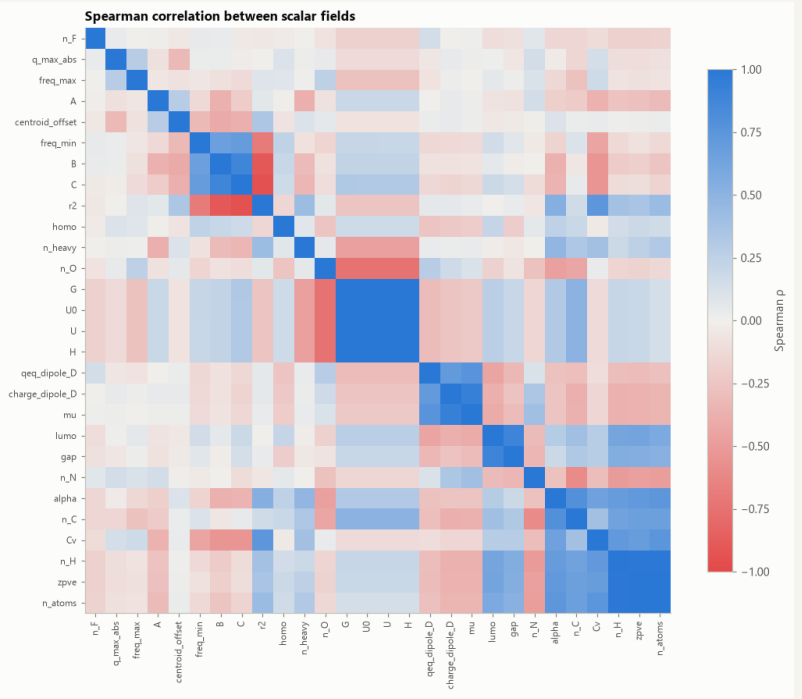


EXPLORATORY DATA ANALYSIS



	role	mean	std	min	1%	50%	99%	max	skew	kurtosis	distinct
A	function of Z and R (headline-legal)	12.046	2,102.153	0.000	1.875	3.092	8.296	619,867.683	272.123	77,723.077	84,827
B	function of Z and R (headline-legal)	1.413	1.817	0.341	0.545	1.376	2.740	437.904	188.568	40,906.854	72,692
C	function of Z and R (headline-legal)	1.133	1.254	0.335	0.515	1.088	2.117	282.945	169.989	33,485.297	64,558
mu	target	2.706	1.487	0.000	0.136	2.522	6.543	29.556	1.489	9.480	44,708
alpha	DFT output (exploration only)	75.162	7.977	6.310	52.630	75.760	93.240	196.620	-0.452	2.294	5,218
homo	DFT output (exploration only)	-0.240	0.022	-0.396	-0.295	-0.241	-0.178	-0.102	0.172	1.682	1,754
lumo	DFT output (exploration only)	0.010	0.046	-0.175	-0.099	0.011	0.091	0.117	-0.172	-0.541	2,376
gap	DFT output (exploration only)	0.251	0.046	0.025	0.152	0.249	0.343	0.505	0.015	-0.583	2,569
r2	DFT output (exploration only)	1,184.099	278.851	19.000	650.699	1,139.860	2,132.826	3,374.753	1.315	3.852	98,588
zpve	DFT output (exploration only)	0.148	0.032	0.016	0.073	0.148	0.228	0.274	-0.010	0.230	59,844
U0	DFT output (exploration only)	-411.276	38.467	-714.568	-496.130	-416.870	-308.468	-40.479	0.309	2.668	98,254
U	DFT output (exploration only)	-411.267	38.467	-714.560	-496.122	-416.862	-308.460	-40.476	0.309	2.668	98,320
H	DFT output (exploration only)	-411.266	38.467	-714.559	-496.121	-416.861	-308.459	-40.475	0.309	2.668	98,287
G	DFT output (exploration only)	-411.309	38.468	-714.602	-496.166	-416.903	-308.499	-40.499	0.309	2.668	98,285
Cv	DFT output (exploration only)	31.470	4.021	6.002	21.802	31.438	40.953	46.961	-0.045	0.475	18,836
n_atoms	function of Z and R (headline-legal)	17.905	2.860	3.000	11.000	18.000	25.000	29.000	-0.058	0.314	26
n_heavy	function of Z and R (headline-legal)	8.806	0.501	1.000	7.000	9.000	9.000	9.000	-3.308	15.787	9
n_C	function of Z and R (headline-legal)	6.378	1.195	0.000	3.000	6.000	9.000	9.000	-0.428	0.329	10
n_H	function of Z and R (headline-legal)	9.098	2.726	0.000	3.000	9.000	16.000	20.000	0.053	0.248	20
n_N	function of Z and R (headline-legal)	0.992	0.990	0.000	0.000	1.000	4.000	7.000	1.158	1.901	8
n_O	function of Z and R (headline-legal)	1.414	0.838	0.000	0.000	1.000	3.000	5.000	0.196	-0.039	6
n_F	function of Z and R (headline-legal)	0.023	0.214	0.000	0.000	0.000	1.000	6.000	11.473	148.853	5
centroid_offset	function of Z and R (headline-legal)	1.595	0.739	0.000	0.341	1.461	3.671	6.648	0.839	0.662	99,197
q_max_abs	DFT output (exploration only)	0.447	0.070	0.058	0.259	0.446	0.608	0.733	-0.312	0.611	80,472
freq_min	DFT output (exploration only)	84.263	50.670	0.166	12.290	72.733	246.143	1,671.422	2.433	24.736	95,714
freq_max	DFT output (exploration only)	3,504.592	264.855	1,412.107	3,107.771	3,515.463	3,841.854	3,907.698	-0.244	-1.210	97,794





Best features: physics estimates of the dipole

QEq dipole

I(Qeq; |μ|)= 0.52

Charge equilibration gives each atom a charge by balancing tabulated electronegativities against Coulomb repulsion on the real geometry, then sums $|\Sigma q_i r_i|$. No fitting at all.

Total bond polarity Σ|Δχ|

I(Σ|Δχ|; |μ|)= 0.38

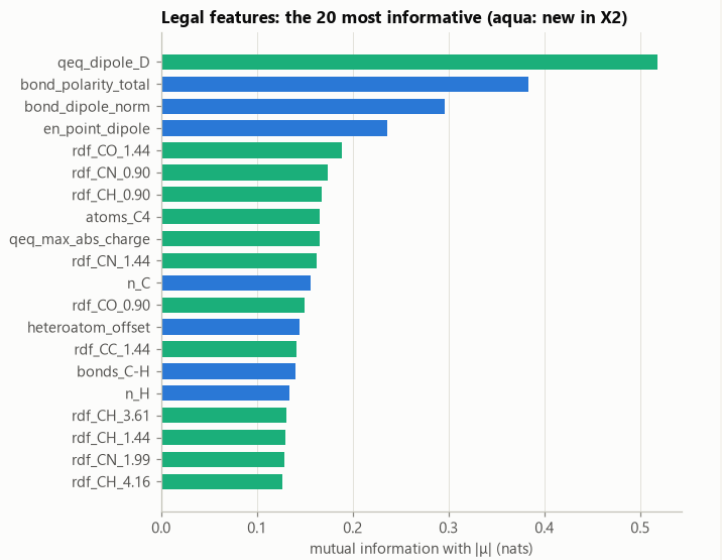
Sum of the magnitude of bond polarity.

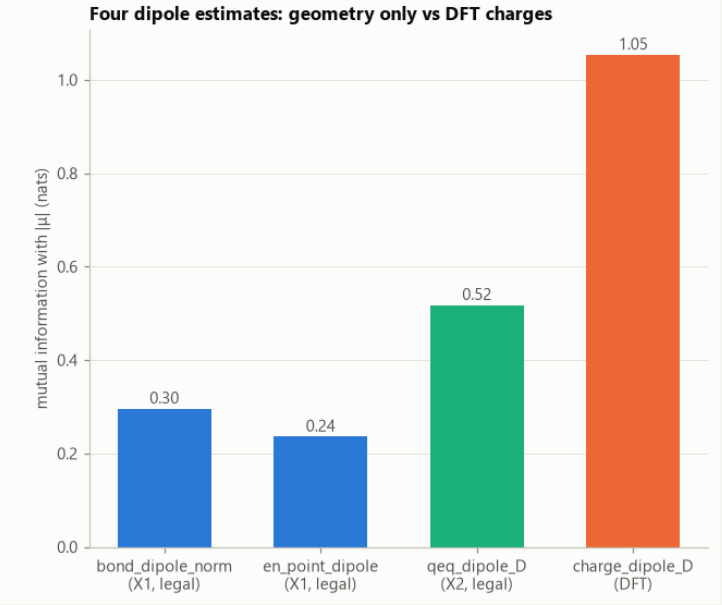
Bond-dipole norm ||Σ Δχ·û||

I(|Σ Δχ·û|; |μ|)= 0.30

Each bond's electronegativity difference, pointed along the bond and added as vectors, so bonds that cancel (as in CO_2) sum to zero.

I(X;Y): Estimated mutual information using sklearn mutual_info_regression. Measured in nats. (0 means independent).





188 legal features from atoms and geometry

5

Composition

Counts of C, H, N, O, F

29

Coulomb-matrix spectrum

Sorted eigenvalues of nuclear repulsion

26

Geometry and bonds

Bond lengths and orders, rings, moments of inertia

19

Functional groups

Including the charged groups of zwitterions

3

QEq dipole

Charge-equilibration estimate: the best single feature

106

Radial distributions

Pair-distance histograms per element pair

- Invariant to rotation, translation and atom relabeling: features change by at most 1e-12, while a raw-coordinate negative control changes by ångströms.
- From 323 total legal features, 135 were removed for low/zero variance, high spearman correlation magnitude, or redundancy.

STANDARDIZATION

Feature scaling

TRACK A - LARGE-DATA CLASSICAL

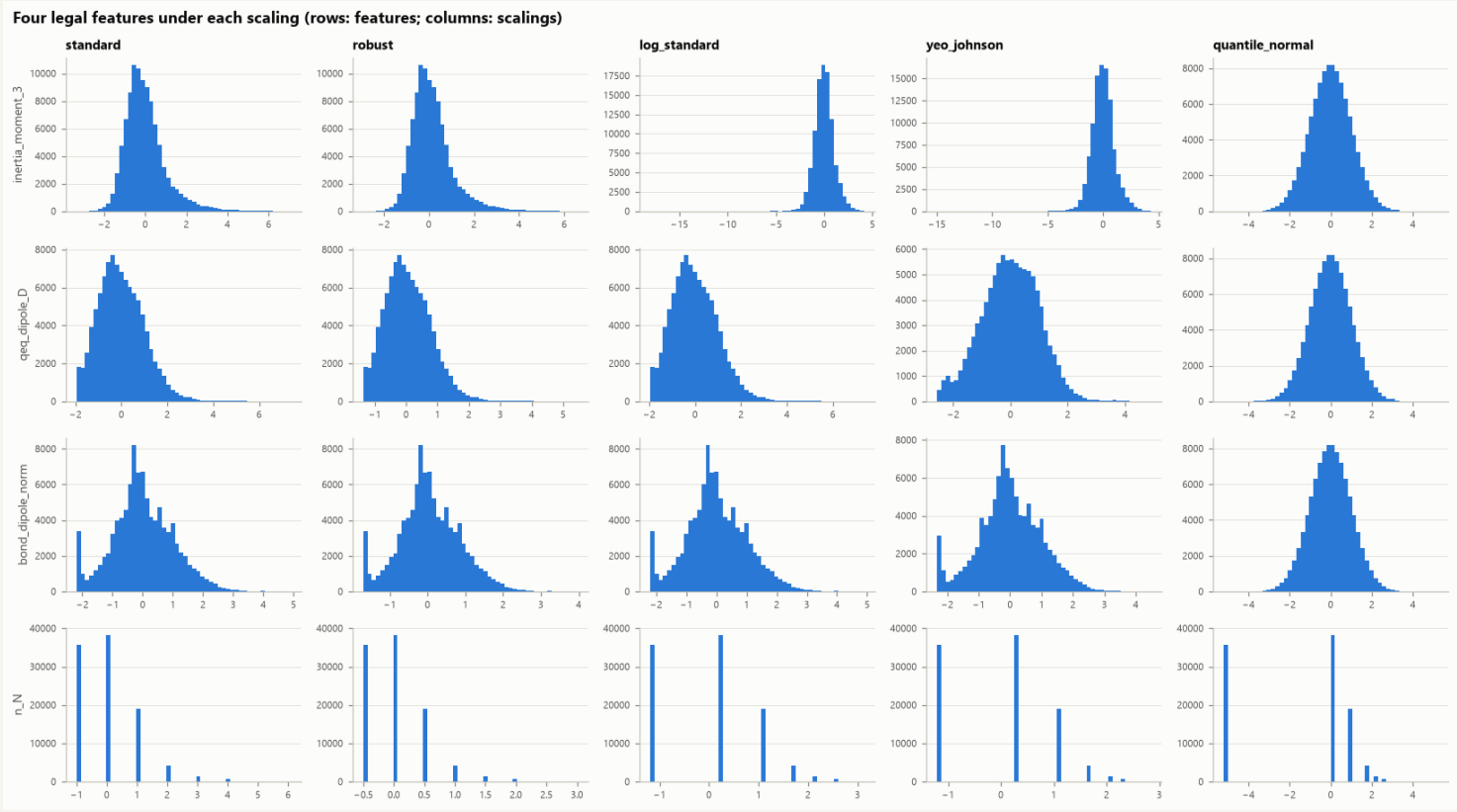
Log-standard + standardized |μ|

Best dev MAE in 6/6 cells: 0.445 D vs 0.452 D with z-scores. Standard target: 0.4208 D vs 0.4226 D for √ target.

TRACK B - QUANTUM COMPARABLE

Yeo–Johnson + √|μ|

Yeo-Johnson won 58% of paired CV cells (0.577 D mean MAE). √ target: 0.5752 D vs 0.5863 D for standard.



scaling	median |skew|	max |skew|	max |value|	share beyond ±3
standard	1.45	58.5	10.0	1.05%
robust	1.40	218.7	10.0	1.40%
log_standard	1.03	58.5	10.0	0.92%
yeo_johnson	0.52	58.5	10.0	0.57%
quantile_normal	1.04	58.6	5.2	46.65%

scaling	Track B: mean CV MAE (D)	Track B: cells won	Track A: mean dev MAE (D)	Track A: cells won
log_standard	0.5802	25%	0.4438	100%
quantile_normal	0.6198	3%	0.4611	0%
robust	0.6146	3%	0.4638	0%
standard	0.5863	11%	0.4516	0%
yeo_johnson	0.5768	58%	0.4560	0%

- “Cell” refers to a combination of SL/ML Model, Feature set, Training size, Random seed.
- For both tracks, multiple differing cell configurations were run for each scaling method. The winning scaling method for a cell was the method yielding the lowest cross validation MAE.
- In this way we experimentally proved that Yeo-Johnson was optimal for these applications.

Compressing 188 features into 10 qubit inputs

One input per qubit, so 188 features had to become 10. We tested three methods with the same model:

PCA

The 10 directions of largest spread; ignores |μ| entirely

0.707 D

Top 10 by mutual information

Keep the 10 single features most informative about |μ|

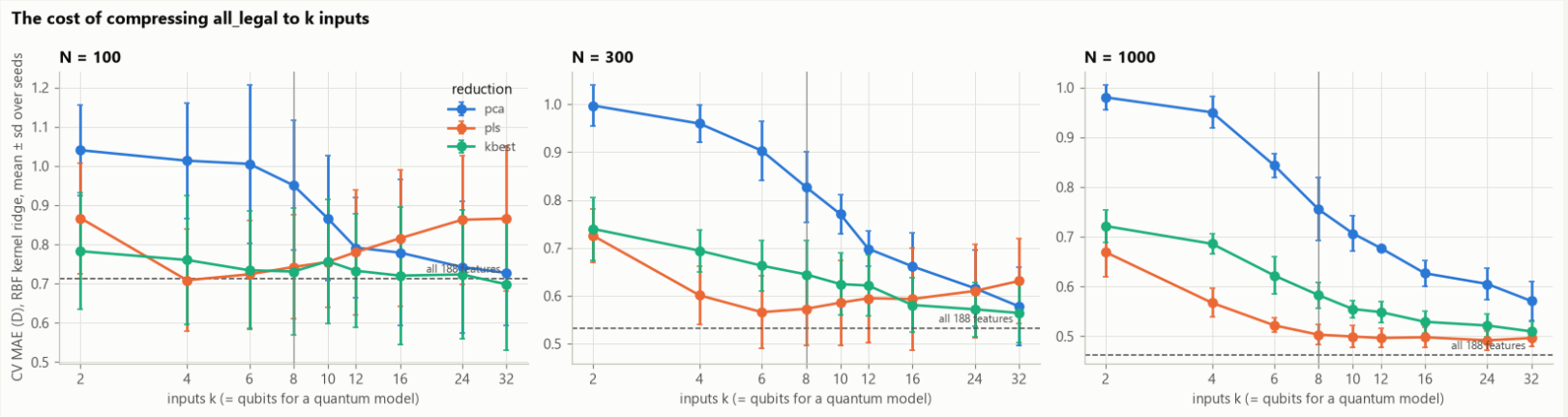
0.555 D

Partial least squares (PLS) · chosen

The 10 directions that vary most together with |μ|

0.499 D

MAE from 5-fold cross-validation, RBF kernel ridge, 1,000 training labels. Why 10: the fewest components within one standard error of the best (24 components, 0.492 D). PLS is supervised and uses the target feature, so it is refit inside every training fold.



mean CV MAE (D) by method, k and N

method	k	n_train 100	300	1000
kbest	2	0.783	0.740	0.722
	4	0.760	0.694	0.686
	6	0.734	0.663	0.622
	8	0.731	0.645	0.583
	10	0.757	0.625	0.555
	12	0.733	0.621	0.549
	16	0.720	0.581	0.529
	24	0.723	0.572	0.522
	32	0.698	0.564	0.510

mean CV MAE (D) by method, k and N

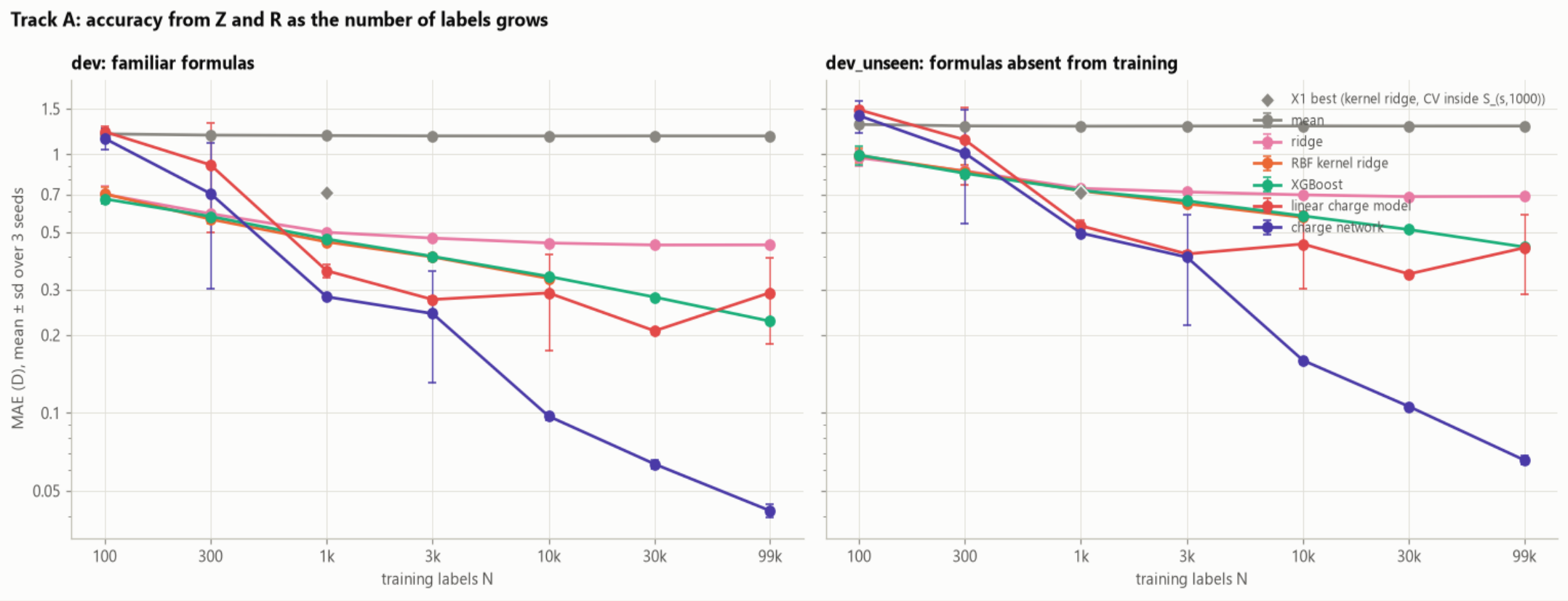
method	k	n_train 100	300	1000
pca	2	1.040	0.997	0.980
	4	1.013	0.960	0.950
	6	1.005	0.903	0.843
	8	0.950	0.827	0.755
	10	0.866	0.771	0.707
	12	0.792	0.698	0.677
	16	0.779	0.661	0.627
	24	0.742	0.616	0.605
	32	0.727	0.578	0.571

mean CV MAE (D) by method, k and N

method	k	n_train 100	300	1000
pls	2	0.867	0.726	0.669
	4	0.709	0.601	0.567
	6	0.724	0.566	0.522
	8	0.742	0.573	0.503
	10	0.755	0.586	0.499
	12	0.780	0.595	0.497
	16	0.816	0.593	0.498
	24	0.863	0.610	0.492
	32	0.866	0.631	0.496

Classical models

To produce the “best” classical model, we created 6 different models using the 188 legal features, averaging their MAE over 3 seeds for each training set.



Models were standard, kernel and RBF kernel ridge, XGBoost, a linear charge model and a charge neural network. (We trained many others but these were the ones that had significant performance)

One encoding circuit, three readouts

Input
Atoms Z and coordinates R

→

Features
188 invariant descriptors

→

Compression
PLS to 10 inputs

→

Qiskit circuit
10-qubit ZZ feature map U(γx)

Fidelity kernel (QKRR)

$k(x, x') = |\langle\psi(x)|\psi(x')\rangle|^2$

1 circuit per pair of molecules. Headline model, chosen by CV.

Projected kernel

RBF on each qubit's ⟨X⟩, ⟨Y⟩, ⟨Z⟩

3 circuits per molecule. After Huang et al. (2021).

⟨Z⟩ feature ridge

one ⟨Z⟩ per qubit, then ridge

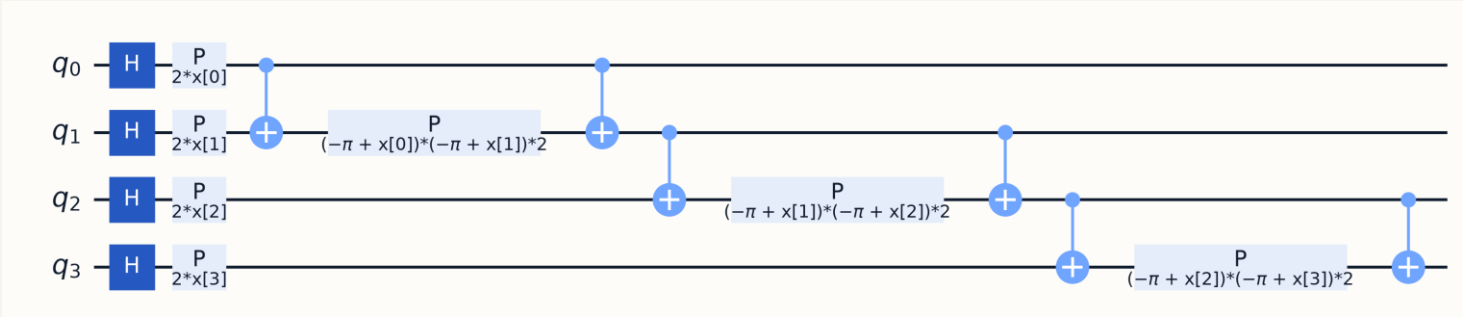
1 circuit per molecule. The team's original model; ran on IBM hardware.

Each readout feeds kernel ridge or ridge regression on $\sqrt{|\mu|}$, mapped back to debye; the angle scale γ and penalty α are tuned by 5-fold CV. Controls: two more encoders, and a product-state kernel with no entangling gates.

Quantum Models

Step 1: Turning samples into quantum states

Input	Features	Compression	Angles	Circuit
Atoms Z, coordinates R	188 numbers per molecule	Scaling + PLS: 10 numbers	× angle scale γ (0.020)	10 qubits, 2 passes



One pass of the ZZ encoder, drawn for 4 of the 10 qubits. x[i] = γ × input i.

The pass runs twice on 10 qubits (36 two-qubit gates). The output is a quantum state of 1,024 amplitudes.

H

Equal mix of 0 and 1 on every qubit

P(2·x[i])

Turns a qubit's phase by an amount set by one input

⊕ – P – ⊕

Entangling block: adds a phase set by two neighbouring inputs together

Step 2: three ways to read the state out

Fidelity kernel (QKRR)

$k(x, x') = |\langle\psi(x)|\psi(x')\rangle|^2$

MEASURES

How much two molecules' states overlap: run U(x′), then U(x) reversed, and count all-zero readouts

THEN

Kernel ridge regression on the table of similarities between training molecules

CIRCUITS

1 per pair of molecules

Test MAE at 1,000 labels: **0.492 D**

Projected kernel

30 measured numbers + RBF kernel

MEASURES

⟨X⟩, ⟨Y⟩ and ⟨Z⟩ of every qubit: 30 numbers per molecule

THEN

A classical RBF kernel on those 30 numbers, then kernel ridge regression

CIRCUITS

3 per molecule

Test MAE at 1,000 labels: **0.493 D**

⟨Z⟩ feature ridge

10 measured numbers + ridge

MEASURES

⟨Z⟩ of every qubit: 10 numbers per molecule, all from one circuit

THEN

Linear ridge regression on those 10 numbers

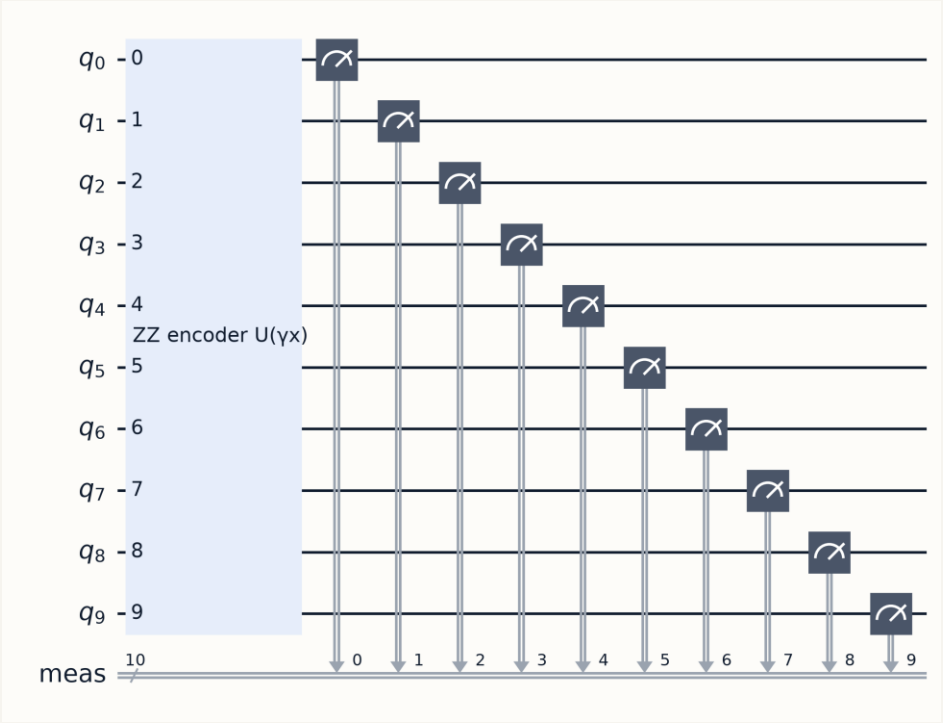
CIRCUITS

1 per molecule

Test MAE at 1,000 labels: **0.545 D**

All three predict √|μ| and square it back to debye. The angle scale γ and ridge penalty α are chosen by 5-fold cross-validation inside the training set, the same way as for the classical twin.

The ⟨Z⟩ feature ridge, step by step



The circuit sent to ibm_quebec: the 10-qubit encoder, then all 10 qubits measured.

1 Encode

The molecule's 10 numbers, times γ, become the rotation angles.

2 Measure

Every qubit reads 0 or 1. Repeat 1,000 times.

3 Average

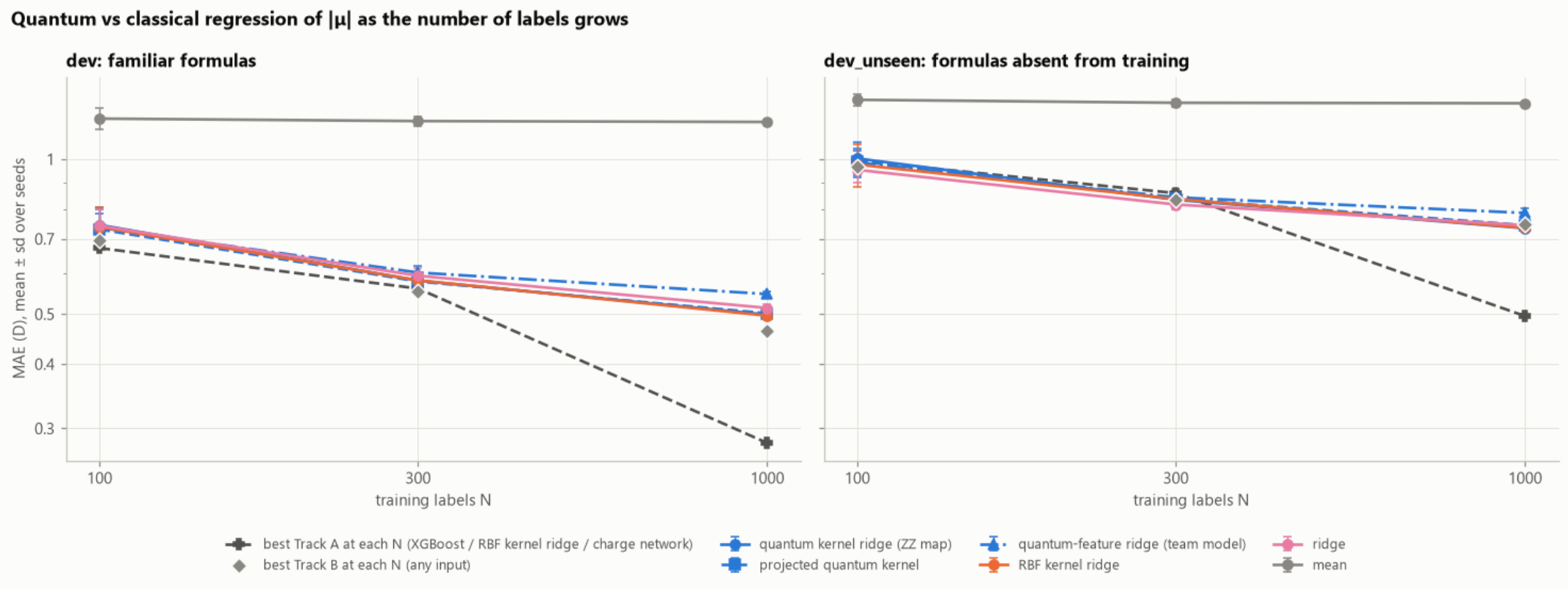
⟨Z⟩ = (zeros − ones) ÷ 1,000 for each qubit: 10 numbers per molecule.

4 Predict

$\sqrt{|\mu|} = b + w_1\langle Z_1\rangle + \ldots + w_{10}\langle Z_{10}\rangle$, squared back to debye. The 11 numbers b and w are fitted on training molecules.

Example: qubit 3 of one molecule read 0 on 700 shots and 1 on 300, so $\langle Z_3\rangle = (700 - 300) \div 1{,}000 = 0.40$.

Quantum Vs Classical



Simulated model performance over 3 seeds

WHY THEY TIE

Entanglement made no measurable difference.

0.490 D

A product-state kernel with no entangling gates, vs 0.492 D for the ZZ map (N = 1,000)

71.4 vs 71.7

Effective parameters of the quantum kernel and of RBF kernel ridge, at the settings CV chose

≈ 0

Mean off-diagonal kernel entry on 16 qubits at γ = 1: at larger angles every molecule looks orthogonal

Cross-validation picks the smallest angle scale, where the fidelity kernel behaves like a wide classical RBF kernel. That is the regime where it works, and in that regime it is effectively classical.

MEASURING PERFORMANCE

Isolating molecule representation choices

Relevant representations/inputs to measure:

01 – Composition-only

- Count of 5 atoms (C, H, N, O, F)
- **5 qubits**

02 – Geometry-aware

- All 188 descriptors, compressed to 10
- **10 qubits**

MEASURING PERFORMANCE

The effect of geometry on training size

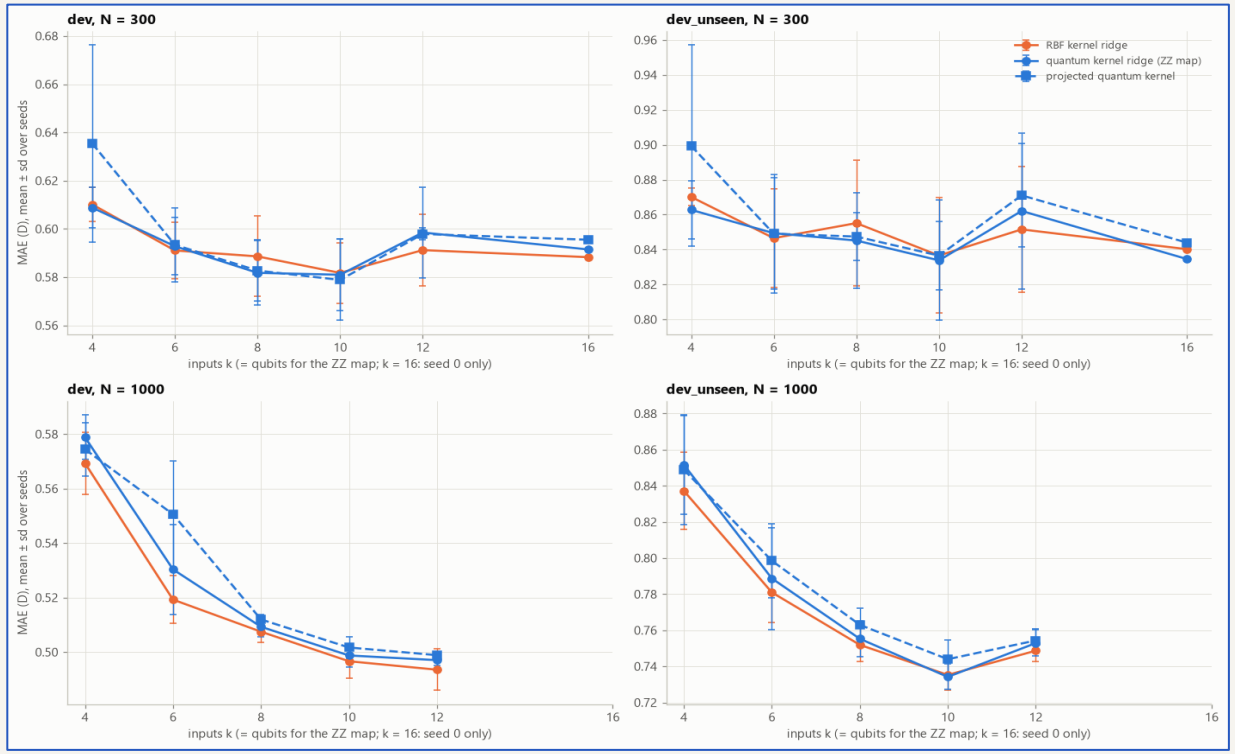
Measuring MAE for composition-only versus geometry-aware

	n_train	100	300	1000
features	model			
composition	projected quantum kernel	**1.050**	**1.018**	**0.985**
	quantum kernel ridge (ZZ map)	1.037	1.025	0.986
	RBF kernel ridge	1.025	1.016	0.986
kbest10	projected quantum kernel	**0.731**	**0.622**	**0.575**
	quantum kernel ridge (ZZ map)	0.718	0.620	0.568
	RBF kernel ridge	0.681	0.612	0.551
pca10	projected quantum kernel	**0.879**	**0.790**	**0.706**
	quantum kernel ridge (ZZ map)	0.806	0.784	0.706
	RBF kernel ridge	0.838	0.770	0.698
pls10	projected quantum kernel	**0.731**	**0.579**	**0.502**
	quantum kernel ridge (ZZ map)	0.745	0.581	0.499
	RBF kernel ridge	0.738	0.582	0.497

MEASURING PERFORMANCE

Validation—the effect of extra qubits

- **Geometry-aware inputs appear to out-perform composition-only inputs**
- **Validate: does the number of qubits (k) impact performance?**
  - **Test k = 4, 6, 8, 10, 12, 16**



Measuring MAE across different k qubits

	n_train	100	300	1000
model	k			
projected quantum kernel	**4**	**0.705**	**0.635**	**0.574**
	6	0.705	0.593	0.551
	8	0.722	0.583	0.512
	10	0.731	0.579	0.502
	12	0.755	0.598	0.499
	16	0.719	0.596	nan
quantum kernel ridge (ZZ map)	**4**	**0.699**	**0.609**	**0.579**
	6	0.698	0.593	0.530
	8	0.726	0.582	0.509
	10	0.745	0.581	0.499
	12	0.755	0.598	0.497
	16	0.699	0.591	nan
RBF kernel ridge	**4**	**0.694**	**0.610**	**0.569**
	6	0.694	0.591	0.519
	8	0.715	0.589	0.507
	10	0.738	0.582	0.497
	12	0.744	0.591	0.494
	16	0.772	0.588	nan

MODELLING

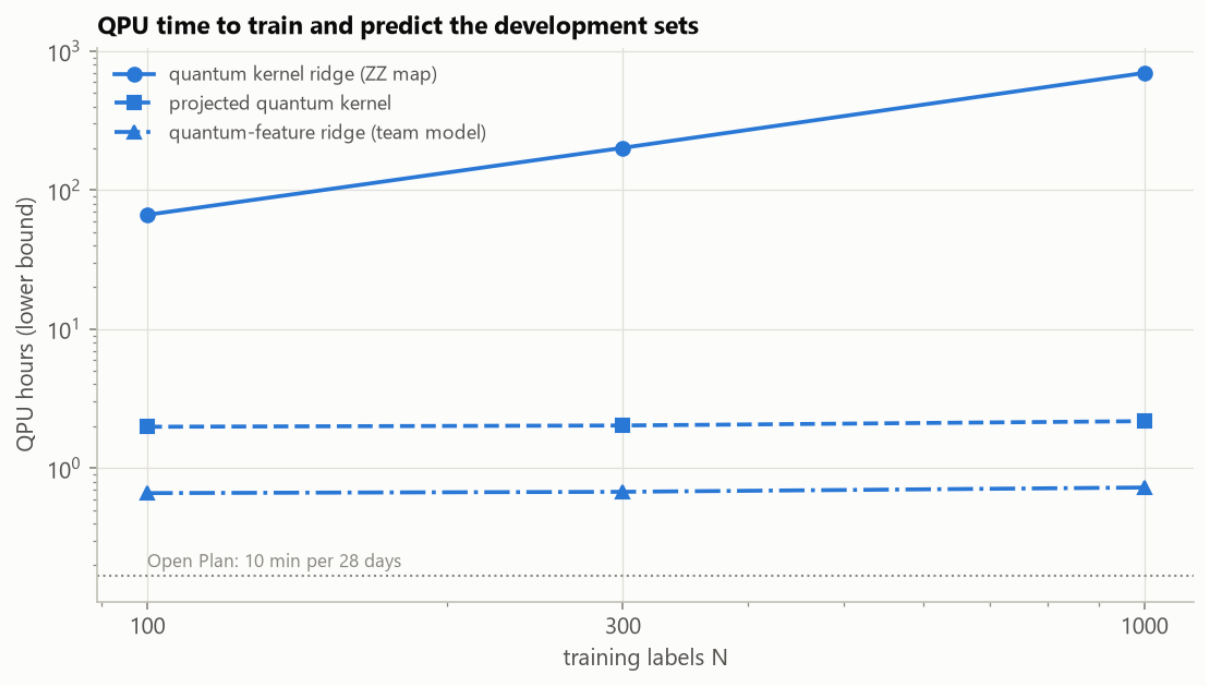
The quantum models tie their classical twin

Model (PLS 10)	N = 100	N = 300	N = 1,000	Unseen, N = 1,000
QKRR, ZZ map	0.735 ± 0.062	0.578 ± 0.011	0.492 ± 0.008	0.437 ± 0.010
Product-state kernel	0.732 ± 0.075	0.578 ± 0.011	0.490 ± 0.009	0.433 ± 0.009
Projected quantum kernel	0.724 ± 0.057	0.578 ± 0.013	0.493 ± 0.007	0.440 ± 0.016
⟨Z⟩ feature ridge	0.731 ± 0.072	0.597 ± 0.010	0.545 ± 0.011	0.493 ± 0.021
RBF kernel ridge	0.730 ± 0.075	0.578 ± 0.010	0.490 ± 0.009	0.433 ± 0.010
RBF kernel ridge, all 188 features	0.662 ± 0.040	0.549 ± 0.012	0.453 ± 0.009	0.420 ± 0.015
Predicting the mean	1.156	1.142	1.138	1.254

Test MAE in debye, mean ± sd over 3 seeds. Paired difference, quantum - twin, at N = 1,000: **+0.001 ± 0.001 D** on familiar formulas and +0.004 ± 0.001 D on unseen ones.

COST, SHOTS AND NOISE

What it would cost on IBM hardware



Lower bound on QPU hours to train and predict the 9,183 development molecules at 1,000 shots per circuit. Dotted line: IBM's free Open Plan, 10 minutes per 28 days.

To train on 1,000 molecules and predict both test sets (22,302 molecules), at 1,000 shots per circuit:

Fidelity kernel (QKRR)
499,500 circuits to train
≈ 1,640 h

Projected kernel
3,000 circuits to train
≈ 5 h

⟨Z⟩ feature ridge
1,000 circuits to train
≈ 1.7 h

Compiled for IBM hardware, one kernel entry is a circuit of depth 206 with 70 two-qubit gates; the encoder alone is depth 109 with 36.

Running our model on IBM hardware

Model
⟨Z⟩ feature ridge, one circuit per molecule

→

Rehearsal
Same circuits on a FakeQuebec noise model

→

Device job
160 circuits × 1,000 shots on ibm_quebec

→

Readout
Ridge refit on the device's measurements

160
circuits: 100 training and 60 test molecules, 10 qubits each

45 s
of QPU time charged (41 s estimated), after about 85 minutes in the queue

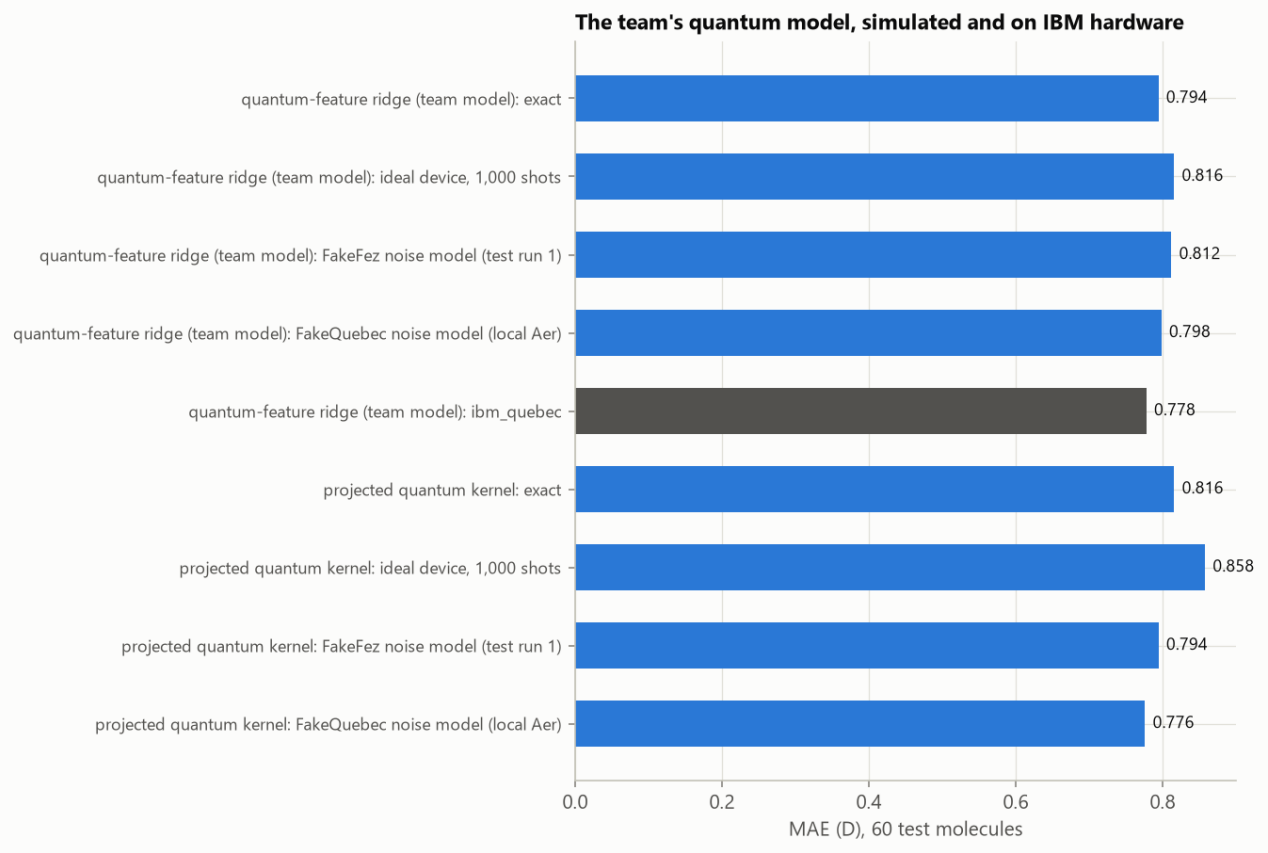
1,000
shots per circuit, with no error mitigation

On the quantum processor: only the 160 circuits.

On our laptop: feature scaling, PLS, the ridge fit and conversion to debye.

IBM job db3da0kvf2bc73csuk60, qiskit-ibm-runtime SamplerV2, accessed through PINQ², tested on 30 familiar and 30 unseen-formula molecules

Hardware matched exact simulation



MAE on the same 60 test molecules, trained on 100. Dark bar: the real device.

0.778 D

⟨Z⟩ feature ridge on the real device, ibm_quebec

0.794 D

The same model in exact simulation

With 60 test molecules, each score is uncertain by ±0.1 D, the device about as well as the simulator.

Model training took 45 seconds on ibm_Quebec

IBM job db3da0kvf2bc73csuk60 - raw shot outcomes in results/hardware/

CONCLUSIONS

Answers to our four questions

01 How fast does error fall as labels grow?

Any 10-input model reaches around 0.5 D with 1,000 labels. While optimized classical networks can reach 0.095 D at 10,000 and 0.040 D at 99,198.

02 Can a quantum model match its classical twin?

Yes, under some circumstances. Every quantum kernel ties RBF kernel ridge within 0.005 D, and a circuit with no entanglement ties too.

03 Do models work on formulas never seen in training?

Yes. On the test set, unseen-formula error was 0.88–0.98× the familiar error for every model. How hard it is depends on which formulas are held out.

04 Does it survive shots, noise and real hardware?

We found that at least shot-aware or feature-based models do. Our ⟨Z⟩ ridge ran on ibm_quebec as accurately as in exact simulation.

NEXT STEPS

What we would do if we had more time:

- Try to train and test a more complex model on quantum hardware
- Explore other types of Quantum models and architectures like splines, ensembles, trees
- Explore different quantum encodings and feature sets
- Explore Chemistry to understand features and their correlations more intuitively for the sake of designing more robust models and features

THANK YOU!